Curriculum Vitae
Mahrud Sayrafi

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Employment

2026 – 2029 **University of Nebraska, Lincoln** — NSF RTG Postdoctoral Fellow
Research Group: Commutative Algebra RTG group

2025 – 2026 **McMaster University** — Britton Postdoctoral Fellow
Research Group: Combinatorial Algebraic Geometry

2025 Spring **Fields Institute for Research in Mathematical Sciences** — Postdoctoral Fellow
Thematic Program: Commutative Algebra and Applications

2024 Fall **Max Planck Institute for Mathematics in the Sciences** — Postdoctoral Fellow
Research Group: Nonlinear Algebra

Education

2024 Ph.D. **University of Minnesota, Twin Cities** — Advisor: Christine Berkesch
Thesis: Diagonalization, Direct Summands, and Resolutions of the Diagonal

2017 B.A. **University of California, Berkeley** (with Honors) — Advisor: David Eisenbud
Thesis: Local Computations in `Macaulay2`

2016 **Math in Moscow** (semester abroad at the Independent University of Moscow)

Awards

2023 – 2024 Doctoral Dissertation Fellowship, University of Minnesota
2020 Honorable Mention, Graduate Research Fellowships Program, NSF
2018 – 2020 John Ordway Fellowship, School of Mathematics, University of Minnesota

Grants

2025 NSF DMS-2508868 for Macaulay2 Workshop at UW-Madison
2023 NSF DMS-2302476 for M2Week at the University of Minnesota
2022 NSF DMS-2206872 for GradMoCCA at the University of Minnesota
2018 DRP Network Seed Grant

Publications

2026 10. "Characterizing multigraded regularity and virtual resolutions on products of projective spaces"
J. Bruce, and L. Cranton Heller. To appear in *Advances in Mathematics.* `arXiv:2110.10705`

2025 9. "Computing Direct Sum Decompositions"
D. Mallory. *Journal of Symbolic Computation* **133**, March (2026). `arXiv:2412.19799`

2024 8. "A short resolution of diagonal for smooth projective toric varieties of Picard rank 2"
M. K. Brown. *Algebra & Number Theory* **18-10** (2024), 1923-1943. `arXiv:2208.00562`

2020 7. "The virtual resolutions package for Macaulay2" `arXiv:1905.07022`
A. Almousa, J. Bruce, and M. C. Loper. *J. of Software for Algebra & Geometry* **10** (2020), 51–60.

2017 6. "What is the optimal way to prepare a Bell state using measurement and feedback?"
L. Martin and K. B. Whaley. *Quantum Sci. & Technol.* **2** (2017), no. 4. `arXiv:1704.00332`

Preprints

2025 6. "Seshadri regions and the asymptotic shape of multigraded regularity" `arXiv:2512.05289`
J. Bruce, and L. Cranton Heller, A. Seceleanu.

2025 5. "Computing global Ext for complexes" `arXiv:2509.25103`
M. Brown, S. Dey, G. Li. Submitted.

2025 4. "Connection Matrices in Macaulay2" `arXiv:2504.01362`
P. Görlach, J. Koefler, A.-L. Sattelberger, H. Schroeder, N. Weiss, F. Zaffalon. Submitted.

2025 3. "Splitting of vector bundles on toric varieties" `arXiv:2412.19793`
Submitted.

2022 2. "Bounds on multigraded regularity" `arXiv:2208.11115`
J. Bruce, and L. Cranton Heller. Submitted.

Invited Talks (selected)

Conferences

2026	Jan.	Joint Math Meetings	Washington, D.C.
2025	Sep.	Route 81 Mathematics Conference	Queen's University, Canada
2024	July	Effective Methods in Algebraic Geometry	MPI MiS, Leipzig, Germany
	May	Spring Western AMS Sectional Meeting	San Francisco State University
	Jan.	Joint Math Meetings	San Francisco, California
2023	Oct.	Fall Central AMS Sectional Meeting	Creighton University, Omaha
	Jul.	SIAM Applied Algebraic Geometry Conference	TU Eindhoven, the Netherlands
	Jul.	Géométrie Algébrique en Liberté (GAeL)	University of Warwick, UK
	Apr.	CA+ Conference	University of Minnesota, Twin Cities
	Mar.	Spring Southeastern AMS Sectional Meeting	Georgia Institute of Technology, Atlanta
2022	Oct.	Fall Western AMS Sectional Meeting	University of Utah, Salt Lake City
	Sep.	Fall Central AMS Sectional Meeting	University of Texas, El Paso
	May	Macaulay2 Conference	Cleveland State University
2021	Apr.	Spring Central AMS Sectional Meeting	University of Cincinnati (virtual)
2019	Aug.	SIAM Applied Algebraic Geometry Conference	Universität Bern, Switzerland
2018	Jan.	Joint Mathematics Meetings	San Diego, California

Seminars

2025	Oct.	Algebra Seminar	Syracuse University
	Oct.	Algebra and Algebraic Geometry Seminar	McMaster University
	Jun.	Commutative Algebra and Applications Seminar	Fields Institute
2024	Dec.	Discrete Geometry Seminar	Technische Universität Berlin
	Dec.	Algebra Seminar	Universität Osnabrück
	Dec.	BIREP Seminar	Universität Bielefeld
	Nov.	EDGE Geometry Seminar	University of Edinburgh, Hodge Institute
	Oct.	Algebra Seminar	Friedrich-Schiller-Universität Jena
	May	Algebra Seminar	University of Oregon, Eugene
	Apr.	Commutative Algebra and Algebraic Geometry Seminar	University of California, Berkeley
	Mar.	Commutative Algebra Seminar	University of Michigan, Ann Arbor
	Feb.	Algebra Seminar	Georgia Tech, Atlanta
	Feb.	Combinatorics Seminar	Washington University in St. Louis
2023	Nov.	Algebraic Geometry and Commutative Algebra Seminar	University of Notre Dame
	Apr.	Commutative Algebra Seminar	University of Nebraska - Lincoln
	Feb.	Algebra and Combinatorics Seminar	Texas A&M University
	Jan.	Commutative Algebra Seminar	University of Illinois, Chicago
2022	Oct.	Algebraic Geometry Seminar	University of Utah
	Oct.	Number Theory and Algebra Seminar	Arizona State University
2021	Oct.	Algebra Seminar	Georgia Tech, Atlanta
	Oct.	Algebra Seminar	Auburn University
	Feb.	Grad/Postdoc Seminar (virtual)	ICERM Combinatorial Algebraic Geometry Program

Posters

2024	Oct.	Group actions, combinatorics, and Fano varieties conference	Universität Innsbruck, Austria
	Sep.	Syzygies and Hilbert Schemes Conference	Jagiellonian University, Poland
	Jan.	SLMath Connections Workshop: Commutative Algebra	SLMath, Berkeley
2023	Nov.	Western Algebraic Geometry Symposium	Washington University in St. Louis
	May	MSRI/SLMath Summer School in Commutative Algebra	University of Notre Dame
2022	Nov.	Western Algebraic Geometry Symposium	University of California, Riverside
	Jun.	Pan-American School in Commutative Algebra	CIMAT, Mexico
2017	Aug.	SIAM Applied Algebraic Geometry Conference	Georgia Tech, Atlanta
2015	Feb.	Southwest Quantum Information and Technology Workshop (SQuInT)	UC Berkeley

Teaching and Mentoring Experience

Department of Mathematics & Statistics, McMaster University

2026 Winter MATH 1MM3: instructor of record for Applied Calculus (class of 340 students).
2025 Fall MATH 3GR3: instructor of record for Abstract Algebra (class of 35 students).

School of Mathematics, University of Minnesota, Twin Cities

2023 Spring MATH 1272: instructor of record for Calculus II (class of 100 students).
2022 Fall MATH 8253: graded homework for Algebraic Geometry (2nd year graduate course).
2022 Spring MATH 1272: instructor of record for Calculus II (class of 90 students).
2019 – 2021 MATH 2243: TA for Linear Algebra & Differential Equations (six sections total).
2018 Fall MATH 1271: TA for Calculus I (two sections total).

2020 Summer REU: mentored P. Cranford, A. Peng, and V. Srinivasan towards `arXiv:2106.12667`.
2019 & 2022 DRP: mentored five undergraduate students in weekly reading projects.

Department of Mathematics, UC Berkeley

2015 Spring Math113: graded homework for Abstract Algebra.
2014 Fall Math116: graded homework and assisted the instructor for Mathematical Cryptography.

Math Center, Irvine Valley College

2013 – 2014 Tutored lower-div courses including linear algebra, differential equations, and discrete math.

Mathobotix, Irvine, CA

2012 – 2014 Developed curriculum for computer science-based problem-solving using Python.

Conference and Workshop Organization

2025 Dec. **CA+AG Session at the Canadian Math Society Meeting** Toronto, Canada
2025 Jul. **Macaulay2 Workshop** Madison, Wisconsin
2024 Nov. **Macaulay2 in the Sciences Workshop** Leipzig, Germany
2023 Jun. **Macaulay2 Workshop & Mini-school** Minneapolis, Minnesota
2022 May **Graduate Meeting on Combinatorial Commutative Algebra**
Weekend graduate event with 12 speakers bringing 70 students and postdocs to Minneapolis.

Mathematical Service

2018 – 2023 **Directed Reading Program**, University of Minnesota, Co-founder and co-organizer
Matched over 250 undergraduate and graduate students in guided mathematics reading projects.
2019 – 2022 **Student Commutative Algebra Meeting**, University of Minnesota, Co-organizer
Held a weekly meetup of friendly neighborhood commutative algebra students.
2021 Apr. **Graduate Student Combinatorics Conference**, Session Chair
Chaired the session on Combinatorial Algebraic Geometry.
2017 – 2018 **Pauline Sperry Undergraduate Lecture Series**, UC Berkeley, Organizer
Inaugurated an annual lecture aimed at providing a role model for marginalized students in math.

Referee Work:
Journal of Pure and Applied Algebra, Journal of Software for Algebra and Geometry

Broader Outreach

2019 – 2023 **Vincent Hall Thespians, University of Minnesota**, Performer
Helped with first-time teaching assistant orientation through situational comedy.
2023 Jan. **Minnesota Project in Mathematics**, Counselor
Helped with workshops and mentored two undergraduate projects during the week-long program.
2019 – 2020 **AMS Graduate Student Blog**, Staff Writer
Wrote about finding community through mathematical art and history.
2018 Jul. **Girls Who Code @ Cloudflare**, Workshop Leader
Instilled an appreciation for mathematics in high school girls using elliptic cryptography puzzles.
2016 – 2017 **Mathematics Undergraduate Student Association**, UC Berkeley, President
Helped build an inclusive and diverse community among undergraduate math students at Berkeley.
2015 & 2018 **Berkeley Splash**, Activity Leader
Led interactive activities about cryptography and elliptic curves for high school students.
2014 & 2015 **Berkeley mini Math Tournament**, Grader and Lecturer
Instilled an appreciation for knot theory in advanced elementary and middle school students.

Workshops and Summer Schools (selected)

2023 Sep. **Syzygies and mirror symmetry**, American Institute of Mathematics, Pasadena, California
Topics: resolutions of the diagonal for toric varieties, homological mirror symmetry

Jul. **Géométrie Algébrique en Liberté**, University of Warwick, UK
Topics: Mori Dream Spaces and quiver GIT, klt singularities, the geometry of curves

Jun. **MRC Derived Categories, Arithmetic and Geometry**, AMS
Topics: Frobenius pushforwards and F-thickness of the blowup $\mathrm{Bl}_5\mathbf{P}^2$

May **MSRI/SLMath Summer School in Commutative Algebra**, University of Notre Dame
TA for mini-course on the geometry of nonstandard syzygies by Daniel Erman

2022 Dec. **RTG Workshop on Birational Complexity**, SCGP, Stony Brook University
Topics: rationality, curves in algebraic varieties, the Cremona group, measures of irrationality

Jun. **Pan-American School in Commutative Algebra**, CIMAT, Mexico
Topics: positive characteristic methods, toric varieties, DG algebras, modules of differentials

2021 Spr. **Combinatorial Algebraic Geometry**, ICERM
Topics: Schubert varieties, toric varieties, tropical varieties, cluster algebras and varieties

2018 May **RTG Summer School in Commutative Algebra**, University of Utah
Topics: limits in positive characteristic, symbolic powers, differential operators, and syzygies.

Mar. **Geometry of Redistricting Workshop**, University of San Francisco
Topics: gerrymandering, voting rights, discrete geometry and graph theory

Open Source Development

Since 2017 **Macaulay2 Internals**
I have contributed to various internal components of Macaulay2, including the engine, interpreter, core mathematical routines, and documentation. I have also contributed to the following packages:

Since 2023 **DirectSummands**: for decomposing modules and coherent sheaves, with D. Mallory.

Since 2023 **Varieties**: for computations involving projective varieties, including complexes of coherent sheaves and their morphisms, with D. Mallory, R. Ramkumar, G. Smith, K. VandeBogert.

Since 2020 **NormalToricVarieties**: added support for pullbacks of coherent sheaves over toric maps, and computations on the Cox ring of toric varieties whose class group has torsion.

Since 2021 **Truncations**: added support for truncations of modules with respect to arbitrary cones, for instance on simplicial toric varieties where the nef and effective cones differ.

Since 2020 **Saturation**: improved and added to core routines for computing annihilators, saturations, and quotients of ideals and modules, with J. Chen and M. Stillman.

Since 2019 **FGLM**: for computing Gröbner bases of zero-dimensional ideals, with D. Peifer.

Since 2018 **VirtualResolutions**: see paper above, with A. Almousa, J. Bruce, and M. Loper.

Since 2017 **LocalRings**: for symbolic computations over local rings, with M. Stillman.

Other Work & Research Experience

2018 Summer **Cloudflare, Inc.** Cryptography Engineering Intern
Launched products involving Tor, Keyless SSL, and distributed randomness generation.

2017 Winter **Proton Research, Inc.** Cryptography Research and Development Intern
Launched support for elliptic curve cryptography in OpenPGP.js.

2017 Summer **Institute for Quantum Computing**, University of Waterloo
Topics: quantum error correcting codes from algebraic curves under John Watrous.

2015 – 2016 **Berkeley Quantum Information and Computation Center**, UC Berkeley
Topics: control theory & entanglement generation under Leigh Martin and Birgitta Whaley.

2014 Summer **SURF @ Institute for Quantum Information and Matter**, Caltech
Topics: quantum game theory and semi-definite optimization under Thomas Vidick.

2013 Summer **SURF @ Jet Propulsion Laboratory**, NASA
Topics: secure multiparty computation and secret sharing systems under Ed Chow.

References

Christine Berkesch	David Eisenbud	Michael Stillman	Craig Westerland
`cberkesc@umn.edu`	`de@berkeley.edu`	`mes15@cornell.edu`	`cwesterl@umn.edu`
Advisor			Teaching